Combinatorics – 1

Counting and bijections

Homework

14 сентября 2026 года

Theoretical minimum. Ordered k-tuples of distinct elements of an n-set number $n(n-1)\cdots(n-k+1)$, subsets $\binom{n}{k}$, multisets $\binom{n+k-1}{k}$ by stars and bars, and ordered set partitions with prescribed block sizes $\frac{n!}{k_1!\cdots k_m!}$. An identity between two counts should be proved by a bijection or by counting one set in two ways; the second method gives $\sum_v \deg v = 2|E|$, $k\binom{n}{k} = n\binom{n-1}{k-1}$ and the hockey-stick identity. Inclusion–exclusion says $|\bigcup A_i| = \sum_{I\neq\varnothing}(-1)^{|I|+1}|\bigcap_{i\in I} A_i|$, because an element lying in exactly m of the sets is counted $\sum_j (-1)^{j+1}\binom{m}{j} = 1$ times; its standard consequences are the derangement count $D_n = n!\sum_{j\leqslant n}(-1)^j/j!$ with $D_n/n! \to e^{-1}$, the number of surjections, and Euler's φ. Counting up to a group action is Burnside's lemma (Abstract algebra – 4).

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1. The 30 edges of a regular icosahedron are labelled $1,\dots,30$. In how many ways can each edge be painted red, white or blue so that each of the 20 triangular faces has two edges of one colour and the third of a different colour? (Putnam 2017)

2. Find the number of ordered 64-tuples $(x_0, x_1, \dots, x_{63})$ of distinct elements of $\{1, 2, \dots, 2017\}$ such that $x_0 + x_1 + 2x_2 + 3x_3 + \cdots + 63x_{63}$ is divisible by 2017. (Putnam 2017)

3. How many positive integers N satisfy all of: N is divisible by 2020; N has at most 2020 decimal digits; the decimal digits of N are a string of consecutive ones followed by a string of consecutive zeros? (Putnam 2020)

4. Find the number of positive integers x with $x < 10^{2006}$ such that $x^2 - x$ is divisible by 10^{2006}. (The related IMC question about unit vectors pairwise orthogonal in triples is Linear algebra – 6, problem 2.) (IMC 2006)

5. Let n be a positive integer. Find the number of pairs P, Q of polynomials with real coefficients such that $P(X)^2 + Q(X)^2 = X^{2n} + 1$ and $\deg P > \deg Q$. (Putnam)

6. Consider an $m \times n$ grid of unit squares indexed by (i, j). There are $(m-1)(n-1)$ coins, placed initially in the squares with $i \leqslant m-1$ and $j \leqslant n-1$. If a coin occupies (i, j) with $i \leqslant m-1$, $j \leqslant n-1$ and the squares $(i+1, j)$, $(i, j+1)$, $(i+1, j+1)$ are empty, it may slide from (i, j) to $(i+1, j+1)$. How many distinct configurations can be reached from the initial one? (Putnam 2023)

7. Define a *growing spiral* to be a sequence of lattice points $P_0 = (0, 0), P_1, \ldots, P_n$ with $n \geqslant 2$ such that the directed segments $P_0P_1, \ldots, P_{n-1}P_n$ point east, north, west, south, east, ... in this cyclic order and their lengths are strictly increasing. How many of the points (x, y) with integer coordinates $0 \leqslant x \leqslant 2011$, $0 \leqslant y \leqslant 2011$ are not the endpoint of any growing spiral? (Putnam)

8. The integers $1, 2, \ldots, 3k+1$ are written in a uniformly random order. What is the probability that at no moment is the sum of the numbers written so far a positive integer divisible by 3? (Putnam 2007)

9. Prove that the number of surjections from an n-set onto a k-set equals $k!\, S(n, k) = \sum_j (-1)^j \binom{k}{j} (k-j)^n$, where $S(n, k)$ is the Stirling number of the second kind (the numbers themselves are studied in Combinatorics – 3). Deduce a formula for the number of ways to partition an n-set into exactly k non-empty blocks.

10. For each positive integer n let $f(n)$ be the number of ways to make $n!$ cents from an unordered collection of coins, each worth $k!$ cents for some $1 \leqslant k \leqslant n$. Prove that for some constant C independent of n, $n^{n^2/2 - Cn} e^{-n^2/4} \leqslant f(n) \leqslant n^{n^2/2 + Cn} e^{-n^2/4}$. (Putnam)